



DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY
2025/2026 University Examination
FIRST SEMESTER OF SECOND YEAR,
BACHELOR OF SCIENCE IN MECHATRONIC ENGINEERING
EMT 3102: SOLID & STRUCTURAL MECHANICS I

DATE: 13th August 2025

2-4pm

DURATION: Ⓢ 2 hours

Instructions:

1. This examination contains 70 marks in total.
2. Question 1 is compulsory. Attempt one question in SECTION A and one question in SECTION B.
3. Show all derivations, diagrams, and calculations clearly.
4. State all assumptions made and include units in final answers.

Question 1 (Compulsory) (30 marks)

- (a) Define the following terms and explain their relevance in mechatronics engineering:
- (i) Stress and strain in structural elements [2 marks]
 - (ii) Modulus of elasticity and its significance in material selection [2 marks]
 - (iii) Factor of safety in mechanical design [2 marks]
- (b) State Hooke's law and explain its limitations. Discuss how understanding material behavior is crucial for designing mechatronic systems components such as robotic arms, sensors, and actuators. [6 marks]
- (c) Derive the general equation for stress in a tapered bar under axial loading. State any assumptions made in the derivation. [6 marks]
- (d) Figure 1 shows a tapered bar with circular cross section, subjected to an axial tensile load $W = 50 \text{ kN}$. The bar tapers linearly from diameter $D_1 = 80 \text{ mm}$ at the fixed end to diameter $D_2 = 40 \text{ mm}$ at the loaded end over a length $L = 1.2 \text{ m}$. Take $E = 200 \text{ GPa}$.

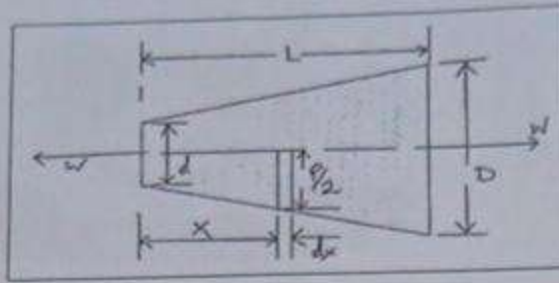


Figure 1: Tapered circular bar under axial loading.

- (i) Calculate the maximum stress in the bar [4 marks]
- (ii) Determine the total extension of the bar [4 marks]
- (iii) Find the stress at the mid-length of the bar [4 marks]

SECTION A: Attempt Only One Question (20 marks)

Question 2: (20 marks)

A compound bar consists of a brass core of diameter $d_b = 30$ mm surrounded by a steel sleeve of internal diameter 30 mm and external diameter $d_s = 50$ mm. The bar is 800 mm long and is subjected to an axial compressive load of $P = 150$ kN at room temperature.

- (a) Derive the general expressions for stress distribution in the brass core (σ_b) and steel sleeve (σ_s) using equilibrium and compatibility conditions. [6 marks]
- (b) Calculate the actual stresses in both materials given that $E_{\text{steel}} = 200$ GPa and $E_{\text{brass}} = 100$ GPa. [4 marks]
- (c) If the temperature of the entire assembly is now raised by $\Delta T = 80^\circ\text{C}$, determine the new stress distribution. Take $\alpha_{\text{steel}} = 12 \times 10^{-6}/^\circ\text{C}$ and $\alpha_{\text{brass}} = 18 \times 10^{-6}/^\circ\text{C}$. [6 marks]
- (d) Calculate the total change in length of the compound bar due to both mechanical loading and thermal effects. [4 marks]

Question 3: (20 marks)

A steel rod of length $L = 2$ m and uniform circular cross-section of diameter $d = 25$ mm is rigidly fixed between two immovable supports.

- (a) If the rod is heated uniformly by $\Delta T = 60^\circ\text{C}$, determine:
 - (i) The thermal stress developed in the rod [3 marks]
 - (ii) The force exerted on the supports [3 marks]

Take $E = 200$ GPa and $\alpha = 12 \times 10^{-6}/^\circ\text{C}$ for steel.

- (b) A tapered steel bar of length $L = 1.5$ m varies linearly in diameter from $d_1 = 40$ mm at one end to $d_2 = 20$ mm at the other end. The bar is rigidly fixed at both ends and subjected to a uniform temperature rise of $\Delta T = 50^\circ\text{C}$.

- (i) Derive an expression for the thermal stress at any cross-section [8 marks]
- (ii) Determine the location and magnitude of maximum thermal stress [6 marks]

Use the same material properties as in part (a).

SECTION B: Attempt Only One Question (20 marks)

Question 4: (20 marks)

- (a) A solid circular shaft of diameter 115 mm is subjected to a torque of 30 kNm. The angle of twist over a length of 4 m is observed to be 0.1 rad. Determine the modulus of rigidity of the material. [4 marks]
- (b) A hollow circular shaft has an external diameter of 125 mm and the internal diameter is three-fourths of the external diameter. If the shear stress on a fiber is 37 MPa due to the applied torque, T :
 - (i) Calculate the applied torque T [3 marks]
 - (ii) Determine the maximum shear stress [3 marks]
 - (iii) Calculate the angle of twist per unit length if $G = 80$ GPa [3 marks]
- (c) A solid circular shaft has a diameter of 85 mm. Find the maximum shear stress and the twist angle over a length of 2.5 m when the shaft is subjected to a torque of 15 kNm. Take $G = 85$ GPa. [4 marks]
- (d) Compare the torsional strength and weight of a solid shaft with that of a hollow shaft having the same external diameter but internal diameter equal to half the external diameter. [3 marks]

Question 5: (20 marks)

- (a) A cylindrical pressure vessel 2.5 m long and 515 mm in diameter with plates 15 mm thick is subjected to an internal pressure of 3.5 MPa. Taking $E = 200$ GPa and Poisson's ratio $\nu = 0.3$ for the vessel material:
 - (i) Calculate the circumferential and longitudinal strains [4 marks]
 - (ii) Determine the original volume and change in volume of the vessel [4 marks]
- (b) A cylindrical thin drum 815 mm in diameter and 4.5 m long is made of 15 mm thick plates. The drum is subjected to an internal pressure of 2.5 MPa. Taking $E = 200$ GPa and Poisson's ratio $\nu = 0.25$, calculate the change in diameter and length of the drum. [4 marks]
- (c) A cylindrical shell of diameter 510 mm is required to withstand an internal pressure of 5 MPa. Find the minimum thickness of the shell, if the maximum tensile strength of the plate material is 410 MPa, the efficiency of the joints is 70%, and the factor of safety is 5. [4 marks]
- (d) A cylindrical shell of 1.5 m diameter is made up of 20 mm thick plates. It is subjected to an internal pressure of 2.5 MPa. Taking the efficiency of the joints to be 75%, calculate the circumferential and longitudinal stresses in the shell. [4 marks]